

Lecture 2: Iteration Complexity Analysis for Block Coordinate Descent Type Method

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The Main Content of the Talk

- **Question:** What is the iteration complexity of the Block Coordinate Descent (BCD) method?
- **Answer:** Scales sublinearly as $\mathcal{O}(1/r)$
 - 1 For general convex problems
 - 2 For various coordinate update rules
 - 3 For a popular family of subproblem solution methods
- **Question:** How does the rate depend on the problem dimension?
- **Answer:** Scales independently, linearly, quadratically, etc, requires case-by-case study

Outline

- 1 Introduction of BCD
 - The Algorithm and Applications
 - The Prior Art
- 2 Analyzing the BCD-type Algorithm
 - The BSUM method and its iteration complexity analysis
 - Special Case: $K = 1$
 - Special Case: $K = 2$
- 3 Analyzing the BCM Algorithm
- 4 Sharpening the Bounds on K

The Problem

- We consider the following problem (with K block variables)

$$\begin{aligned} \text{minimize} \quad & f(x) := g(x_1, \dots, x_K) + \sum_{k=1}^K h_k(x_k) \\ \text{subject to} \quad & x_k \in X_k, \quad k = 1, \dots, K \end{aligned} \quad (\text{P})$$

- $g(\cdot)$ smooth convex function; $h_k(\cdot)$ nonsmooth convex function;
- $x = (x_1^T, \dots, x_K^T)^T \in \mathbb{R}^n$ is a partition of x ; $X_k \subseteq \mathbb{R}^{n_k}$

Applications

- Lots of applications in practice
- One of the most well-known application is the LASSO problem

$$\min_x \frac{1}{2} \|Ax - b\|^2 + \lambda \|x\|_1$$

- Each scalar $x_k \in \mathbb{R}$ is a block variable

$$\min_x \frac{1}{2} \left\| \sum_{k=1}^K A_k x_k - b \right\|^2 + \lambda \sum_{k=1}^K |x_k|$$

Applications (cont.)

- Rate maximization in uplink wireless communication network
- K users, a single base station (BS)
- Each user has n_t (n_r) transmit (receive) antennas
- Let $\mathbf{C}_k \in \mathbb{R}^{n_t \times n_t}$ denote user k 's transmit covariance matrix
- $\mathbf{H}_k \in \mathbb{R}^{n_r \times n_t}$ the channel matrix between user k and the BS
- Then the uplink channel capacity optimization problem is

$$\begin{aligned} \min_{\{\mathbf{C}_k\}_{k=1}^K} \quad & -\log \det \left| \sum_{k=1}^K \mathbf{H}_k \mathbf{C}_k \mathbf{H}_k^T + \mathbf{I}_{n_r} \right| \\ \text{s.t.} \quad & \mathbf{C}_k \succeq 0, \text{Tr}[\mathbf{C}_k] \leq P_k, \quad k = 1, \dots, K \end{aligned}$$

- The celebrated **iterative water-filling algorithm (IWFA)** [Yu-Cioffi 04] is simply BCD with cyclic update rule

The Problem

- Let us assume that the gradient of $g(\cdot)$ is block-wise Lipschitz continuous

$$\begin{aligned}\|\nabla_k g([x_{-k}, x_k]) - \nabla_k g([x_{-k}, \hat{x}_k])\| &\leq M_k \|x_k - \hat{x}_k\|, \quad \forall x \in X, \forall k \\ \|\nabla g(x) - \nabla g(z)\| &\leq M \|x - z\|, \quad \forall x, z \in X\end{aligned}$$

- Let $M_{\min} = \min M_k$, $M_{\max} = \max M_k$

The Algorithm

- Consider the cyclic **block coordinate minimization (BCM)**

- At iteration $r + 1$, block k updated by

$$x_k^{r+1} \in \arg \min_{x_k \in X_k} g(x_1^{r+1}, \dots, x_{k-1}^{r+1}, x_k, x_{k+1}^r, \dots, x_K^r) + h_k(x_k)$$

- Sweep over all blocks in cyclic order (a.k.a Gauss-Seidel rule)
- Popular for solving modern large-scale problems
 - Many known variants – **block coordinate descent (BCD)** type
 - Block selection rule:** cyclic, randomized, parallel, greedy,...
 - Block update rule:** exact minimization, proximal gradient,...

The Algorithm (cont.)

- **Variant.** cyclic **block coordinate proximal gradient** (BCPG)

- 1 At iteration $r + 1$, block k updated by

$$x_k^{r+1} = \arg \min_{x_k \in X_k} u_k(x_k; x_1^{r+1}, \dots, x_{k-1}^{r+1}, x_k^r, \dots, x_K^r) + h_k(x_k)$$

- 2 Sweep over all blocks in cyclic order

- $u_k(x_k; y)$: a quadratic approximation of $g(\cdot)$ w.r.t. x_k

$$u_k(x_k; y) = g(y) + \langle \nabla_k g(y), x_k - y_k \rangle + \frac{L_k}{2} \|x_k - y_k\|^2.$$

- If $h \equiv 0$, then the iterates becomes

$$x_k^{r+1} = x_k^r - \frac{1}{L_k} \nabla_k g(x_1^{r+1}, \dots, x_{k-1}^{r+1}, x_k^r, \dots, x_K^r)$$

- Achieving **descent** in each block (L_k large enough)

The Result

- **Question:** How many number of iterations until solving (P) to ϵ -OPT (a.k.a iteration complexity)?
- **Result 1:** BCPG-type inexact scheme achieves $\mathcal{O}(1/r)$ rate (with various coordinate update rules)
- **Result 2:** BCM achieves $\mathcal{O}(1/r)$ rate (with various coordinate update rules)

The Result (cont.)

- **Question:** How does the rate depend on the problem dimension?
- **Answer:** Scales (almost) independently, linearly, etc, requires case-by-case study [Sun-H. 15]

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Prior Art: BCPG

- A large literature on analyzing BCPG-type algorithm
- **Randomized BCPG**: blocks picked randomly
- Strongly convex problems, **linear rate** [Richtárik-Takáč 12]
- Convex problems with **$\mathcal{O}(1/r)$ rate**
 - 1 Smooth [Nesterov 12]
 - 2 Smooth + L1 penalization [Shalev-Shwartz-Tewari 11]
 - 3 General nonsmooth (P) [Richtárik-Takáč 12][Lu-Xiao 13]

Prior Art: BCPG (cont.)

- How about cyclic BCPG? Less is known
- For strongly convex problems, **linear rate**
- For general convex problems, **$\mathcal{O}(1/r)$ rate?**
 - 1 LASSO, when satisfies the so-called “isotonicity assumption” (assumption on the data matrix) [Saha-Tewari 13]
 - 2 Smooth [Beck-Tetruashvili 13]
 - 3 General nonsmooth, i.e., (P)?

Prior Art: BCM

- How about cyclic BCM? Even less is known
- For strongly convex problems, **linear rate**
- For convex problem **$\mathcal{O}(1/r)$ rate**
 - 1 Smooth unconstrained problem with $K = 2$ (two-block variables) [Beck-Tetruashvili 13]
 - 2 Other cases?
- Other coordinate update rules (e.g., Gauss-Southwell)?

The Summary of Results

- A summary of existing results on sublinear rate for BCD-type
- NS=NonSmooth, S=Smooth, C=Constrained, U=Unconstrained, K=K-block, 2=2-Block
- GS=Gauss-Seidel, GSo=Gauss-Southwell, EC=Essentially-Cyclic

Table: Summary of Prior Art

Method	Problem	$\mathcal{O}(1/r)$ Rate
GS-BCPG	S-C-K	$\checkmark$
GS/GSo/EC-BCPG	NS-C-K	?
GS-BCM	S-U-2	$\checkmark$
GS/EC-BCM	NS-C-K	?

This Work

- This work shows the following results [H.-Wang-Razaviyayn-Luo 14]
- NS=NonSmooth, S=Smooth, C=Constrained, U=Unconstrained, K=K-block, 2=2-Block
- GS=Gauss-Seidel, GSo=Gauss-Southwell, EC=Essentially-Cyclic

Table: Summary of Prior Art + This Work

Method	Problem	$\mathcal{O}(1/r)$ Rate
GS-BCPG	S-C-K	✓
GS/GSo/EC-BCPG	NS-C-K	✓
GS-BCM	S-U-2	✓
GS/EC-BCM	NS-C-K	✓

What's Missing?

- Why we care about iteration complexity?
 - ① Characterize how fast the algorithms progresses
 - ② Estimate practical performance for big data problems
- For large scale problems, the scaling with respect to K matters!
- When $K = 10^6$, two algorithms with $\mathcal{O}(K/r)$ and $\mathcal{O}(K^2/r^2)$...

The State-of-the-Art

- The classical gradient method scales $\mathcal{O}(M/r)$ (independent of K)
- In [Saha-Tewari 13], GS-BCPG for LASSO with “isotonicity assumption” scales $\mathcal{O}(M/r)$
- The GS-BCPG for smooth problems [Beck-Tetruashvili 13] scales $\mathcal{O}(K^2 M/r)$ in the worst case
- The analysis in [H.-Wang-Razaviyayn-Luo 14] scales similarly
- Better bounds?

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The BSUM method

- We first introduce the so-called **block successive upper-bound minimization** (BSUM) method [Razaviyayn-H.-Luo 13]
- A general framework includes BCPG/BCM as special cases

- **Main steps**

- 1 At iteration $r + 1$, block k updated by

$$x_k^{r+1} \in \arg \min_{x_k \in X_k} u_k(x_k; x_1^{r+1}, \dots, x_{k-1}^{r+1}, x_{k'}^r, \dots, x_K^r) + h_k(x_k)$$

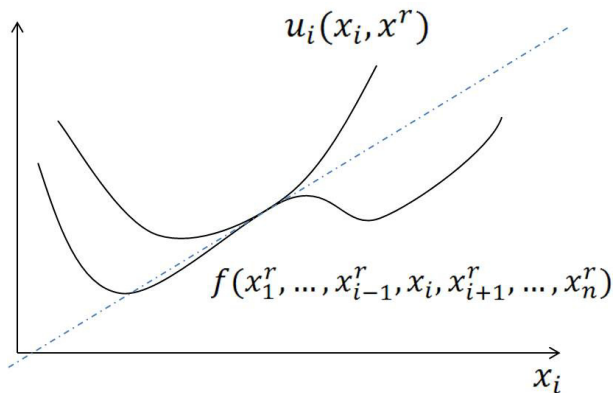
- 2 Sweep over all blocks in cyclic order

- $u_k(x_k; y)$ an approximation of $g(x)$ at point y
- In the following mainly focus on the GS rule

The BSUM method (cont.)

- Assumptions on $u_k(v_k; x)$
 - (a) $u_k(x_k; x) = g(x)$, $\forall x$ [locally tight]
 - (b) $u_k(v_k; x)$ is continuous in v_k and x [continuity]
 - (c) $u_k(v_k; x) \geq g(v_k, x_{-k})$, $\forall v_k, x$ [upper-bound]
 - (d) $u_k(v_k; x)$ **strongly convex** in v_k with coefficient γ_k [strong-convexity]
 - (e) $u_k(v_k; x)$ has Lipchitz gradient with coefficient L_k [Lip-gradient]
 - (f) $\nabla u_k(x_k; x) = \nabla_k g(x_k, x_{-k})$ [gradient-consistent]
- Call $u_k(v_k; x)$ as a **valid upper-bound**

The BSUM method (cont.)



The BSUM method (cont.)

- The convergence of BSUM is guaranteed, even for nonconvex problems [Razaviyayn-H.-Luo 13]
- Reduces to the **GS BCPG** when $u_k(x_k; y)$ is a quadratic approx.
- Reduces to **GS BCM** when no approx. is used
- Many other choices of upper-bound functions [Razaviyayn-H.-Luo 13] [Mairal 13]
- See recent work [H.-Razaviyayn-Luo-Pang 15] for applications of this method in signal processing/communications/machine learning

Iteration Complexity Analysis for BSUM

- Define X^* as the optimal solution set, and let $x^* \in X^*$ be one of the optimal solutions
- Define the **optimality gap** as

$$\Delta^r := f(x^r) - f(x^*)$$

- **Main proof steps**

- 1 **Sufficient Descent:** $\Delta^r - \Delta^{r+1}$ large enough
- 2 **Estimate Cost-To-Go:** Δ^{r+1} small enough
- 3 **Estimate The Rate:** Show $(\Delta^{r+1})^2 \leq c (\Delta^r - \Delta^{r+1})$

Sufficient Descent

Lemma

(Sufficient Descent) For BSUM, we have for all $r \geq 1$

$$\Delta^r - \Delta^{r+1} = f(x^r) - f(x^{r+1}) \geq \gamma \|x^r - x^{r+1}\|^2,$$

where the constant $\gamma := \frac{1}{2} \min_k \gamma_k > 0$.

- Proof easy, requires block strong convexity (BSC) condition
- Measure the descent by the distance traveled

Estimating Cost-To-Go

- Define $R := \max_{x \in X} \max_{x^* \in X^*} \{ \|x - x^*\| : f(x) \leq f(x^1) \}$
- Utilize the optimality of $\{x_k^{r+1}\}$ and the Lip-gradient of g and u_k 's
- Estimate the cost yet to be minimized, again **by the distanced traveled**

Lemma

(Cost-to-go Estimate) For BSUM, we have

$$(\Delta^{r+1})^2 = (f(x^{r+1}) - f(x^*))^2 \leq 2K(M^2 + L_{\max}^2)R^2 \|x^{r+1} - x^r\|^2$$

where $L_{\max} = \max_k L_k$.

$\mathcal{O}(1/r)$ Iteration Complexity

- Linking the above two lemmas, we obtain for all $r \geq 1$

$$\Delta^r - \Delta^{r+1} \geq \frac{\gamma}{2K(M^2 + L_{\max}^2)R^2} (\Delta^{r+1})^2 := \sigma (\Delta^{r+1})^2$$

- By a simple induction argument, we obtain

$$\Delta^r \leq \frac{c}{\sigma} \frac{1}{r}, \text{ with } c := \max\{4\sigma - 2, f(\mathbf{x}^1) - f^*, 2\}.$$

Special Case for cyclic BCPG

- For BCPG, the upper-bound function is given by

$$u_k(x_k; y) := g(y) + \langle \nabla_k g(y), x_k - y_k \rangle + \frac{L_k}{2} \|x_k - y_k\|^2$$

where L_k is the largest eigenvalue of $\nabla_k^2 g(x)$

- Then we have $\gamma_k = L_k = M_k$, and the bound reduces to

$$\Delta^r \leq 8 \frac{cKM}{\min_k M_k} \frac{MR^2}{r}$$

- The bound is in the same order as the one stated in [Beck-Tetruashvili 13, Theorem 6.1]
- In the worst case $M/M_{\min} = \mathcal{O}(K)$, so the red part scales with K^2

Remark

- Same analysis for GS-BCM with **per-block strongly convexity (BSC)**

- 1 LASSO with scalar blocks:

$$\min \left\| \sum_{k=1}^K a_k x_k - b \right\|^2 + \|x\|_1$$

- 2 Rate maximization problem in wireless communication networks

$$\max_{p_k} \log \left(\sum_{k=1}^K |h_k|^2 p_k + 1 \right), \text{ s.t. } p_k \in P_k, \forall k$$

- The analysis extends to other popular update rules

- 1 Gauss-Southwell (i.e., greedy coordinate selection)
- 2 Maximum Block Improvement (MBI) [Chen-Li-He-Zhang 13]
- 3 Essentially cyclic

- Key challenge.** Complexity without BSC?

Special Case: $K = 1$

- Let us briefly consider the case where $K = 1$

$$\begin{aligned} \min \quad & f(x) := g(x) + h(x) \\ \text{s.t.} \quad & x \in X. \end{aligned}$$

- BSUM reduces to the successive upper-bound minimization (SUM)

$$x^{r+1} \in \min_{x \in X} u(x; x^r) + h(x).$$

Special Case: $K = 1$

- Let us make the following assumptions on the function $u(v; x)$.
 - (a) $u(x; x) = g(x)$, $\forall x \in X$. [locally tight]
 - (b) $u(v; x) \geq g(v)$, $\forall v \in X, \forall x \in X$. [upper-bound]
 - (c) $\nabla u(x; x) = \nabla g(x)$, $\forall x \in X$. [gradient consistency]
 - (d) For any given x , $u(v; x)$ has Lipschitz continuous gradient, that is

$$\|\nabla u(v; x) - \nabla u(\hat{v}; x)\| \leq L\|v - \hat{v}\|, \forall \hat{v}, v \in X, \forall x \in X,$$

where $L > 0$ is some constant [Lip-continuous]

- Main difference.** no longer requires that $u(\cdot)$ is strongly convex

The $\mathcal{O}(1/r)$ Iteration Complexity

Lemma

(Lipschitz constant estimate) Suppose $g(x)$ is convex, and $u(v; x)$ satisfies the previous assumption. Then we must have

$$\|\nabla g(v) - \nabla g(x)\| \leq L\|v - x\|, \quad \forall x, v \in X. \quad (2.1)$$

That is, ∇g is Lipschitz continuous with the coefficient no larger than L .

- Our main result is that the SUM algorithm converges sublinearly **without BSC**, i.e.,

$$\Delta^{r+1} \leq \frac{c}{\sigma} \frac{1}{r}, \quad \text{with} \quad \sigma = \frac{1}{32R^2L}$$

Application 1: Iterative Reweighted Least Squares (IRLS)

- The IRLS is popular for solving sparse recovery and Fermat-Weber problem [Beck 13]; Consider the following problem

$$\min_x h(x) + \sum_{j=1}^{\ell} \|A_j x + b_j\|_2, \quad \text{s.t. } x \in X$$

where $A_j \in \mathbb{R}^{k_i \times m}$, $b_j \in \mathbb{R}^{k_i}$, $h(x)$ is some convex function

- Introduce a constant $\eta > 0$ and consider a smooth approximation:

$$\min_x h(x) + g(x) := h(x) + \sum_{j=1}^{\ell} \sqrt{\|A_j x + b_j\|_2^2 + \eta^2}, \quad \text{s.t. } x \in X.$$

Application 1: Iterative Reweighted Least Squares (IRLS)

- Consider the following problem

$$\min_x h(x) + g(x) := h(x) + \sum_{j=1}^{\ell} \sqrt{\|A_j x + b_j\|_2^2 + \eta^2}, \quad \text{s.t. } x \in X.$$

- The IRLS generates the following iterates

$$x^{r+1} = \arg \min_{x \in X} \left\{ h(x) + \frac{1}{2} \sum_{j=1}^{\ell} \frac{\|A_j x + b_j\|^2 + \eta^2}{\sqrt{\|A_j x^r + b_j\|^2 + \eta^2}} \right\}.$$

Interpretation as a 2-block problem

- IRLS is equivalent to BCM applied to a two-block problem

$$\begin{aligned} \min \quad & h(\mathbf{x}) + \frac{1}{2} \sum_{j=1}^{\ell} \left(\frac{\|A_j \mathbf{x} + b_j\|^2 + \eta^2}{z_j} + z_j \right) \\ \text{s.t.} \quad & \mathbf{x} \in X, \quad z_j \in [\eta/2, \infty), \quad \forall j. \end{aligned}$$

- The both blocks are not necessarily strongly convex
- Utilizing such two-block BCM interpretation, [Beck 13] shows that the IRLS converges sublinearly, when $h(x)$ has Lip-gradient

IRLS as SUM Algorithm

- **A new perspective.** the IRLS is SUM in disguise
- Our simple iteration complexity for SUM can be directly applied
- Let us consider the following function:

$$u(\mathbf{x}; \mathbf{x}^r) = \frac{1}{2} \sum_{j=1}^{\ell} \left(\frac{\|A_j \mathbf{x} + b_j\|^2 + \eta^2}{\sqrt{\|A_j \mathbf{x}^r + b_j\|^2 + \eta^2}} + \sqrt{\|A_j \mathbf{x}^r + b_j\|^2 + \eta^2} \right).$$

- Easy to verify that this function is an upper bound (Ass. [a]–[d]) for

$$g(\mathbf{x}) = \sum_{j=1}^{\ell} \sqrt{\|A_j \mathbf{x} + b_j\|_2^2 + \eta^2}$$

IRLS as SUM Algorithm

- It is easy to verify the [local tightness] property $g(x^r) = u(x^r; x^r)$
- To show the [upper bound] property, use the arithmetic-geometric inequality

$$\begin{aligned} u(x; x^r) &= \frac{1}{2} \sum_{j=1}^{\ell} \left(\frac{\|A_j x + b_j\|^2 + \eta^2}{\sqrt{\|A_j x^r + b_j\|^2 + \eta^2}} + \sqrt{\|A_j x^r + b_j\|^2 + \eta^2} \right) \\ &\geq \sum_{j=1}^{\ell} \sqrt{\|A_j x + b_j\|^2 + \eta^2} = g(x), \quad \forall x \in X. \end{aligned}$$

- The rest of the properties are also easy to verify
- So despite lack of strong convexity in $u(\cdot)$, still achieves $\mathcal{O}(1/r)$ convergence

Application 2: A General Pattern for $K = 2$

- The $K = 1$ case can also be used to analyze problems with $K = 2$
- Let us make the following assumptions
 - (a) The problem $\min_{x_2 \in X_2} f(x_1, x_2)$ has a **unique solution** [inner problem]
 - (b) The gradient of $g(x_1, x_2)$ with respect to x_1 is Lipschitz continuous [outer problem]

$$\|\nabla_1 g(x_1, x_2) - \nabla_1 g(v_1, x_2)\| \leq M_1 \|x_1 - v_1\|.$$

- No assumption of BSC; no assumption of the Lip-gradient for x_2
- **Main Result** Applying BSUM to such 2-block problem is **equivalent** to applying SUM to certain **single-block** problem

Application 2: A General Pattern for $K = 2$

- Let us consider the following iterations

$$\begin{aligned} x_2^{r+1} &= \arg \min_{x_2 \in X_2} g(x_2, x_1^r) + h_2(x_2) \\ x_1^{r+1} &\in \arg \min_{x_1 \in X_1} u_1(x_1; x_1^r, x_2^{r+1}) + h_1(x_1). \end{aligned} \tag{2.2}$$

- Inner-problem** solved exactly; **outer-problem** use upper-bound
- Consider an equivalent **single-block** problem (with x_1 as variable)

$$\min_{x_1 \in X_1} \ell(x_1) + h_1(x_1) := \min_{x_1 \in X_1} \min_{x_2 \in X_2} f(x_1, x_2),$$

where we have defined $\ell(x_1) := \min_{x_2 \in X_2} g(x_1, x_2) + h_2(x_2)$

- Denote an optimal solution of the inner problem by the mapping: $x_2^*(x_1) : X_1 \rightarrow X_2$ (which is a singleton by assumption)

Application 2: A General Pattern for $K = 2$

- To proceed the analysis, define a new **single-variable** function

$$u(v_1; x_1) := u_1(v_1; x_1, x_2^*(x_1)) + h_2(x_2^*(x_1)).$$

- **Claim:** for all $x_1, v_1 \in X_1$, $u(v_1; x_1)$ is an upper bound for $\ell(v_1)$
- First we verify the **[local tightness]** assumption

$$\begin{aligned} \ell(x_1) &= g(x_1, x_2^*(x_1)) + h_2(x_2^*(x_1)) \\ &\stackrel{(a)}{=} u_1(x_1; x_1, x_2^*(x_1)) + h_2(x_2^*(x_1)) = u(x_1; x_1) \end{aligned}$$

where (a) because $u_1(x_1; x)$ is an upper bound function for $g(\cdot, x_2)$

Application 2: A General Pattern for $K = 2$

- Then we verify the **[upper-bound]** property; note

$$\begin{aligned}
 u(v_1; x_1) &= u_1(v_1; x_1, x_2^*(x_1)) + h_2(x_2^*(x_1)) \\
 &\geq g(v_1, x_2^*(x_1)) + h_2(x_2^*(x_1)) \\
 &\geq \min_{x_2} g(v_1, x_2) + h_2(x_2).
 \end{aligned} \tag{2.3}$$

- Third, we verify the **[gradient consistency]** assumption; Recall for any given $x_1 \in X_1$, the mapping $x_2^*(x_1)$ is a **singleton**
- From a classical result, we have

$$\nabla \ell(x_1) = \nabla_1 g(x_1, \hat{x}_2), \quad \forall x_1 \in X_1$$

where $\hat{x}_2 = \arg \min_{x_2 \in X_2} f(x_1, x_2)$.

- It follows

$$\begin{aligned}
 \nabla \ell(x_1) &= \nabla_1 g(x_1, \hat{x}_2) = \nabla u_1(x_1; x_1, \hat{x}_2) \\
 &= \nabla u_1(x_1; x_1, x_2^*(x_1)) = \nabla u(x_1; x_1)
 \end{aligned}$$

Application 2: A General Pattern for $K = 2$

- **Conclusion.** The iteration

$$x_2^{r+1} = \arg \min_{x_2 \in X_2} g(x_2, x_1^r) + h_2(x_2)$$

$$x_1^{r+1} \in \arg \min_{x_1 \in X_1} u_1(x_1; x_1^r, x_2^{r+1}) + h_1(x_1).$$

can be viewed as generated by

$$x_1^{r+1} \in \arg \min u(v_1; x_1^r)$$

which computes the optimal solution for the **single-variable** problem

$$\min \ell(x_1) + h(x_1)$$

- A SUM algorithm, sublinear convergence

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Motivating Examples

- **Example 1.** Consider the group-LASSO problem

$$\min_x \left\| \sum_{k=1}^K A_k x_k - b \right\|^2 + \lambda \sum_{k=1}^K \|x_k\|_2$$

- x_k subproblem (semi)closed-form solution; A_k 's can be rank-deficient
- **Example 2.** Consider a rate maximization problem in wireless networks (K user, multiple antenna, etc)

$$\min_{\{C_k\}_{k=1}^K} -\log \det \left| \sum_{k=1}^K H_k C_k H_k^T + I_{n_r} \right|, \quad \text{s.t.} \quad C_k \succeq 0, \text{Tr}[C_k] \leq P_k, \forall k$$

- If H_k is not full row rank, each C_k subproblem is not strongly convex
- **Our previous results do not apply!**

Iteration Complexity for BCM?

- **The Algorithm.** At iteration $r + 1$, update:

$$x_k^{r+1} \in \min_{x_k \in X_k} g \left(x_1^{r+1}, \dots, x_{k-1}^{r+1}, x_k, x_{k+1}^r, \dots, x_K^r \right) + h_k(x_k)$$

- **Key challenges.**
 - 1 No per-block strong convexity (SC) anymore
 - 2 Multiple optimal solutions
 - 3 The sufficient descent estimate is lost (**why?**)

Rate Analysis for GS-BCM (no BSC)

- **Key idea.** Using a different measure to gauge progress.
- **Key steps:** Still three-step approach
 - Sufficient descent

$$\Delta^r - \Delta^{r+1} \geq \frac{1}{2M} \sum_{k=1}^K \|\nabla g(w_k^{r+1}) - \nabla g(w_{k+1}^{r+1})\|^2.$$

where

$$w_k^{r+1} := [x_1^{r+1}, \dots, x_{k-1}^{r+1}, x_k^r, x_{k+1}^r, \dots, x_K^r].$$

- Cost-to-go estimate

$$(\Delta^{r+1})^2 \leq 2K^2 R^2 \sum_{k=1}^K \|\nabla g(w_{k+1}^{r+1}) - \nabla g(w_k^{r+1})\|^2, \forall x^* \in X^*.$$

- Matching the previous two and obtain...

Rate Analysis for cyclic BSUM (no BSC)

Theorem

(H.-Wang-Razaviyayn-Luo 14) Let $\{x^r\}$ be the sequence generated by the BCM algorithm with G-S rule. Then we have

$$\Delta^r = f(x^r) - f^* \leq \frac{c_5}{\sigma_5} \frac{1}{r}, \quad \forall r \geq 1, \quad (3.4)$$

where the constants are given below

$$\begin{aligned} c_5 &= \max\{4\sigma_5 - 2, f(x^1) - f^*, 2\}, \\ \sigma_5 &= \frac{1}{2MK^2R^2}, \end{aligned} \quad (3.5)$$

Outline

- 1 Introduction of BCD
 - The Algorithm and Applications
 - The Prior Art
- 2 Analyzing the BCD-type Algorithm
 - The BSUM method and its iteration complexity analysis
 - Special Case: $K = 1$
 - Special Case: $K = 2$
- 3 Analyzing the BCM Algorithm
- 4 Sharpening the Bounds on K

What's Missing?

- Why we care about iteration complexity?
 - ① Characterize how fast the algorithms progresses
 - ② Estimate practical performance for big data problems
- For large scale problems, the scaling with respect to K matters!
- When $K = 10^6$, two algorithms with $\mathcal{O}(K/r)$ and $\mathcal{O}(K^2/r^2)$...

The State-of-the-Art

- The classical gradient method scales $\mathcal{O}(M/r)$ (independent of K)
- In [Saha-Tewari 13], GS-BCPG for LASSO with “isotonicity assumption” scales $\mathcal{O}(M/r)$
- The GS-BCPG for smooth problems [Beck-Tetruashvili 13] scales $\mathcal{O}(K^2 M/r)$ in the worst case
- The analysis in [H.-Wang-Razaviyayn-Luo 14] scales similarly
- Better bounds?

Sharpening the Bounds on K ?

- All the rates scale **quadratically** in K (in worst case)
- Next we sharpen the bound for the following quadratic problem

$$\min f(x) := \frac{1}{2} \left\| \sum_{k=1}^K A_k x_k - b \right\|^2 + \sum_{k=1}^K h_k(x_k), \quad \text{s.t. } x_k \in X_k, \forall k \quad (\text{Q})$$

Sharpening the Bounds on K ?

- Consider the BCPG algorithm
- Remove a K factor in the worst case
- Matches the complexity of gradient descent (almost independent of K) for some special cases

A Reminder

- Let us assume that the gradient of $g(\cdot)$ is block-wise Lipschitz continuous

$$\begin{aligned}\|\nabla_k g([x_{-k}, x_k]) - \nabla_k g([x_{-k}, \hat{x}_k])\| &\leq M_k \|x_k - \hat{x}_k\|, \quad \forall x \in X, \forall k \\ \|\nabla g(x) - \nabla g(z)\| &\leq M \|x - z\|, \quad \forall x, z \in X\end{aligned}$$

- $u_k(x_k; y)$: a quadratic approximation of $g(\cdot)$ w.r.t. x_k

$$u_k(x_k; y) = g(y) + \langle \nabla_k g(y), x_k - y_k \rangle + \frac{L_k}{2} \|x_k - y_k\|^2.$$

- If $h \equiv 0$, then the iterates becomes

$$x_k^{r+1} = x_k^r - \frac{1}{L_k} \nabla_k g(x_1^{r+1}, \dots, x_{k-1}^{r+1}, x_k^r, \dots, x_K^r)$$

The Result

- **Question:** How does the rate bound depend on K ?
- **Result 1:** BCPG+quadratic g . If $L_k = M_k$, then the rate scales

$$\mathcal{O}\left(\log^2(K) \left(\frac{M_{\max}}{M} + \frac{M}{M_{\min}}\right) MR^2\right)$$

- **Result 2:** BCPG+quadratic g . If $L_k = M$, then the rate scales

$$\mathcal{O}\left(\log^2(K) MR^2\right)$$

- **Result 3:** For problems with $\frac{M_{\max}}{M_{\min}} = \mathcal{O}(1)$, the above rates are tight

Rate Analysis for GS-BCPG

Theorem (Sun-H. 15)

The iteration complexity of using BCPG to solve (Q) is given below

- ① Suppose the stepsizes are chosen as $L_k = M$, $\forall k$, then

$$\Delta^{(r+1)} \leq 3 \max \left\{ \Delta^0, 4 \log^2(2K) M \right\} \frac{R^2}{r+1}.$$

- ② Suppose the stepsizes are chosen according to:

$$L_k = \lambda_{\max}(A_k^T A_k) = M_k, \quad \forall k.$$

Then we have

$$\Delta^{(r+1)} \leq 3 \max \left\{ \Delta^0, 2 \log^2(2K) \left(M_{\max} + \frac{M^2}{M_{\min}} \right) \right\} \frac{R^2}{r+1}$$

Tightness of the Bounds

- We briefly discuss the tightness of the bound over K
- **Question:** Can we improve the bounds further in the order of K ?
- Construct a simple quadratic problem

$$\min \quad g(x) := \left\| \sum_{k=1}^K A_k x_k \right\|^2$$

with

$$A = \begin{bmatrix} 1 & 1 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \cdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 1 & 1 \end{bmatrix}.$$

- Consider scalar blocks, then $M_k = 1$ for all k
- We also have (Note: $M/M_k = \mathcal{O}(1)$)

$$M = \lambda_{\max}(A) = 1 + 2 \cos(i\pi/(K+1)), \quad k = 1, \dots, K,$$

Tightness of the Bounds (cont.)

- We can show that after running a **single pass** of GS-BCD/BCPG

$$\Delta^{(1)} \geq \frac{9(K-3)}{4(K-1)} \|x^{(0)} - x^*\|^2, \forall K \geq 3.$$

- Specialized our bound to this problem predicts that the gap is *at most*

$$\Delta^{(1)} \leq \left(\frac{M - M_{\min}}{M_{\min}} \frac{1}{2} + \frac{1}{4} \right) \|x^{(0)} - x^*\|^2 M \leq 36 \|x^{(0)} - x^*\|^2.$$

- As $K \rightarrow \infty$, the previous two bounds match, up to a constant dimensionless factor $1/16$
- **Conclusion.** The derived bound is tight for the case $M/M_{\min} = \mathcal{O}(1)$

Numerical Comparison

- We compare the performance of GD and BCD over the constructed problem

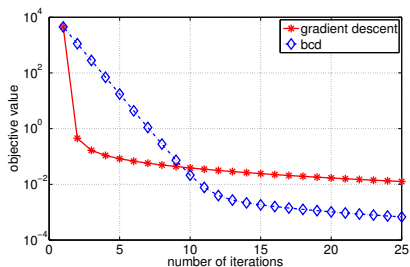
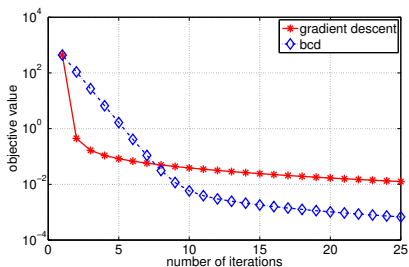


Figure: Comparison of the gradient method and GS-BCD method to solve the constructed. Left, $K = 100$. Right, $K = 1000$

Extensions

The proposed technique also applies to the following scenarios

- Quadratic strongly convex problems (reduces a K factor)
- General nonlinear convex smooth problems

Comparisons

Table: Comparison of Various Iteration Complexity Results

Lip-constant 1/Stepsize	Diag. Hess. $M_i = M$ $L_i = M$	Full Hess. $M_i = \frac{M}{K}$ Large step $L_i = \frac{M}{K}$	Full Hess. $M_i = \frac{M}{K}$ Small step $L_i = M$
GD	M/r	N/A	M/r
R BCGD	M/r	$M/(Kr)$	M/r
GS BCGD	KM/r	K^2M/r	KM/r
GS BCGD (QP)	$\log^2(2K)M/r$	$\log^2(2K)KM/r$	$\log^2(2K)M/r$

Conclusion

- We show the $\mathcal{O}(1/r)$ sublinear rate of convergence for a few BCD-type methods
- We also manage to reduce the dependency of the rates on the problem dimension
- **Observation.** conservative stepsize obtains better theoretical rate bound (but worse practical performance)

Future Work

- Still a gap in the rate bound
- **Question.** Can the GS-BCD/BCPG matches the bound of GD for general convex K -block problems (i.e., independent of problem dimension)? If not, construct an example?
- **Question.** Does random permutation help?

Thank You!

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